AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 1: Brainstorming & Ideation

Objective: Identify the school IT helpdesk problem and define the automation idea for classifying incoming incidents.

1. Problem Identified

- The school IT helpdesk receives multiple incident requests every day from students and teachers.
- Common requests include Wi-Fi/network issues, projector failures, password or login problems, and slow computers.
- IT staff currently review each request manually and select the correct category and subcategory.
- The manual process is time-consuming, error-prone and difficult to scale.

2. Proposed Idea

- Create a no-code ServiceNow solution using Flow Designer.
- Analyze keywords in the incident short description and description fields.
- Automatically identify the issue type and assign the appropriate Category and Subcategory.
- Send an automated email notification to the caller after ticket creation.

3. Initial Classification Idea

- WiFi/Network → Category: Network; Subcategory: Wi-Fi.
- Projector → Category: Hardware; Subcategory: Projector.
- Password/Login → Category: Access; Subcategory: Forgot Password.
- Slow/Hanging → Category: Performance; Subcategory: Slow Computer.

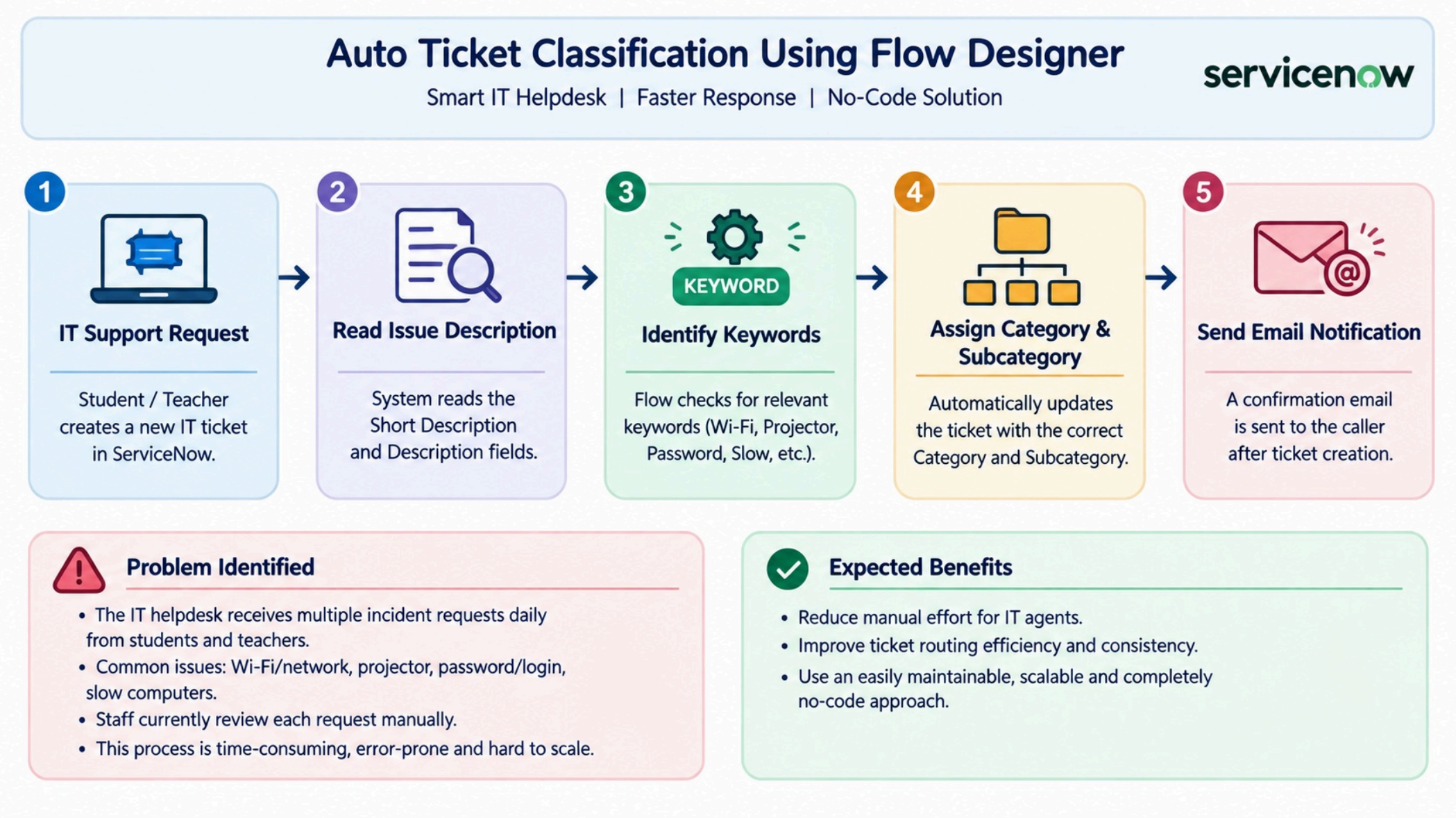
4. Expected Benefit

- Reduce manual effort for IT agents.
- Improve ticket routing efficiency and consistency.
- Use an easily maintainable, scalable and completely no-code approach.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 1 · Illustrative Workflow / Configuration Diagram

Illustration for documentation; use actual ServiceNow screenshots as implementation evidence where required.



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PHASE 2: Requirement Analysis

Objective: Convert the identified helpdesk problem into clear business and functional requirements.

1. Business Requirements

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Send an automated email notification to the caller upon ticket creation.
- Store ticket information in a structured and standardized format.
- Ensure easy maintenance and future scalability.

2. Required Ticket Information

- Number – Auto Number.
- Caller – Reference to sys_user.
- Category – Choice: Network, Hardware, Access, Performance.
- Subcategory – Choice: Wi-Fi, Projector, Forgot Password, Slow Computer.
- Short Description – String.
- Description – String.
- State – Choice: New, In progress, On hold, Resolved, Closed.
- Assigned Group and Assigned to – Reference fields.

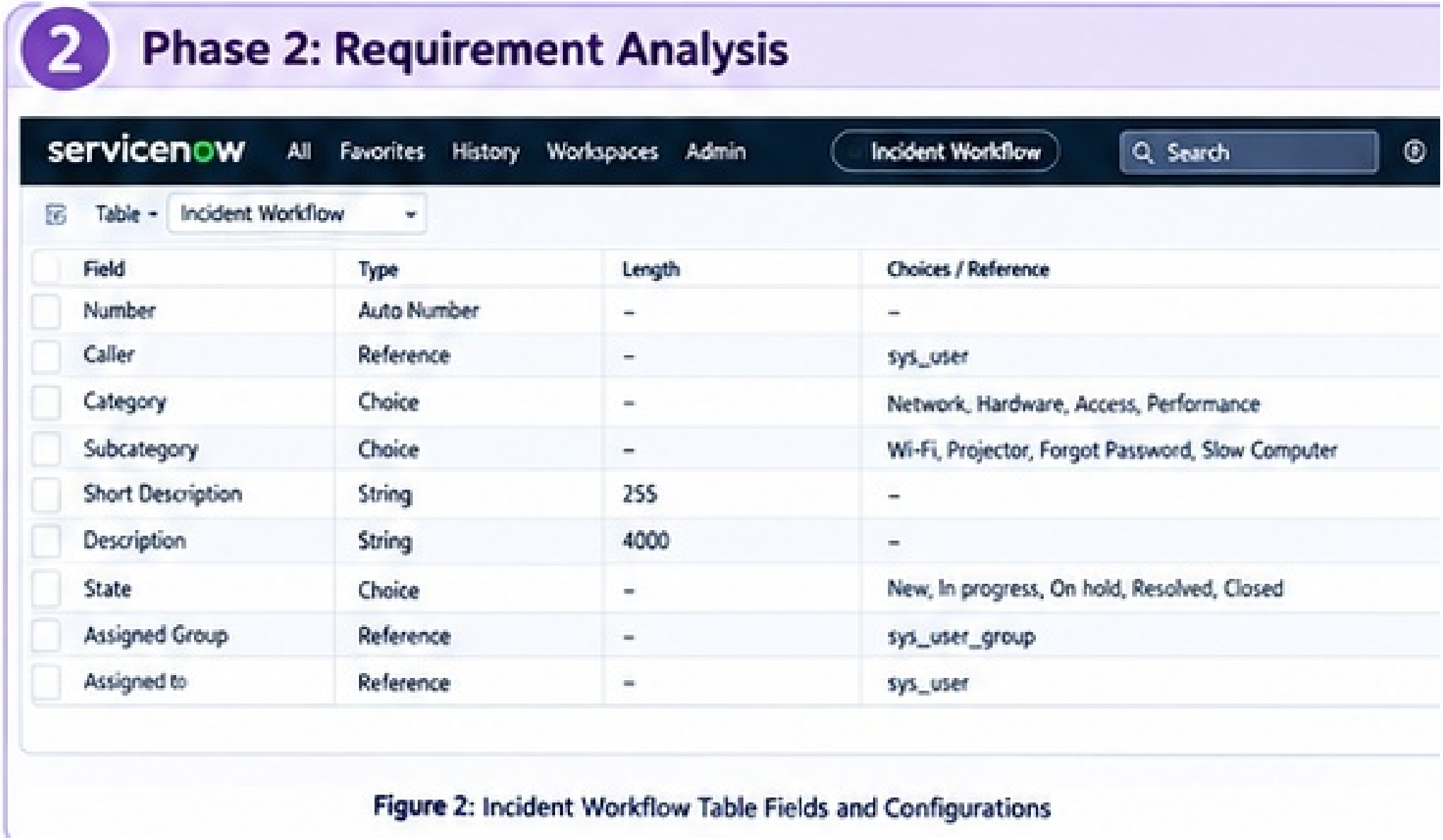
3. Automation Requirements

- The flow should start when a new Incident Workflow record is created.
- The flow should check whether Category is empty.
- Keyword-based conditions should determine the Category and Subcategory.
- After classification, an email should be sent to the caller.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 2 · Illustrative Workflow / Configuration Diagram

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Field	Type	Length	Choices / Reference
Number	Auto Number	–	–
Caller	Reference	–	sys_user
Category	Choice	–	Network, Hardware, Access, Performance
Subcategory	Choice	–	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	255	–
Description	String	4000	–
State	Choice	–	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	–	sys_user_group
Assigned to	Reference	–	sys_user

Figure 2: Incident Workflow Table Fields and Configurations

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PHASE 3: Project Design

Objective: Design the ServiceNow data structure, fields, dependencies and automation flow before implementation.

1. Table Design

- Create a custom table named Incident Workflow to store ticket records.
- Enable Auto Number so each ticket receives a structured unique number.

2. Field Design

- Create fields for Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned to.
- Configure appropriate data types, choices and reference fields.

3. Dependent Choice Design

- Make Subcategory dependent on Category.
- Wi-Fi depends on Network.
- Projector depends on Hardware.
- Forgot Password depends on Access.
- Slow Computer depends on Performance.

4. Flow Design

- Trigger: Record Created.
- Table: Incident Workflow.
- Condition: Category is Empty.
- Use If / Else If logic to inspect Short Description.
- Update the record with Category and Subcategory.
- Send confirmation email to the caller.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 3 · Illustrative Workflow / Configuration Diagram

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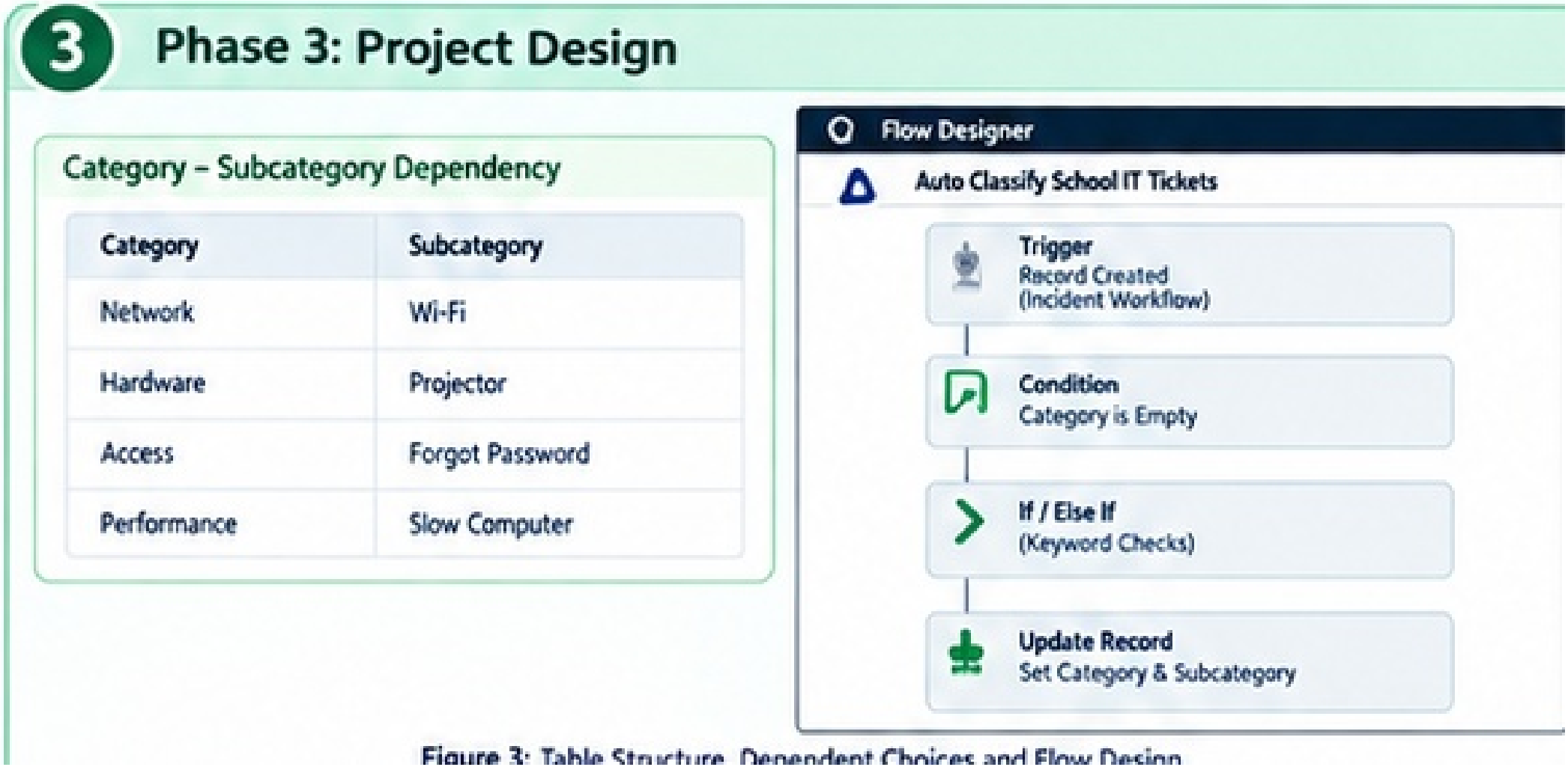


Figure 3: Table Structure, Dependent Choices and Flow Design

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PHASE 4: Project Planning

Objective: Plan the ServiceNow implementation activities, configuration sequence and deliverables.

1. Implementation Plan

- Create a Local Update Set named Project Update Set.
- Set State to In progress and Application to Global.
- Make the update set the current set before configuration.
- Create the Incident Workflow table.
- Create and configure all required fields and choices.
- Configure Category–Subcategory dependency.
- Build and configure the Flow Designer automation.
- Configure email notification.
- Test the complete workflow.
- Complete the update set and export it for reuse.

2. Planned Automation Rules

- WiFi or Network keyword → Network / Wi-Fi.
- Projector keyword → Hardware / Projector.
- Password or Login keyword → Access / Forgot Password.
- Slow or Hanging keyword → Performance / Slow Computer.

3. Planned Deliverables

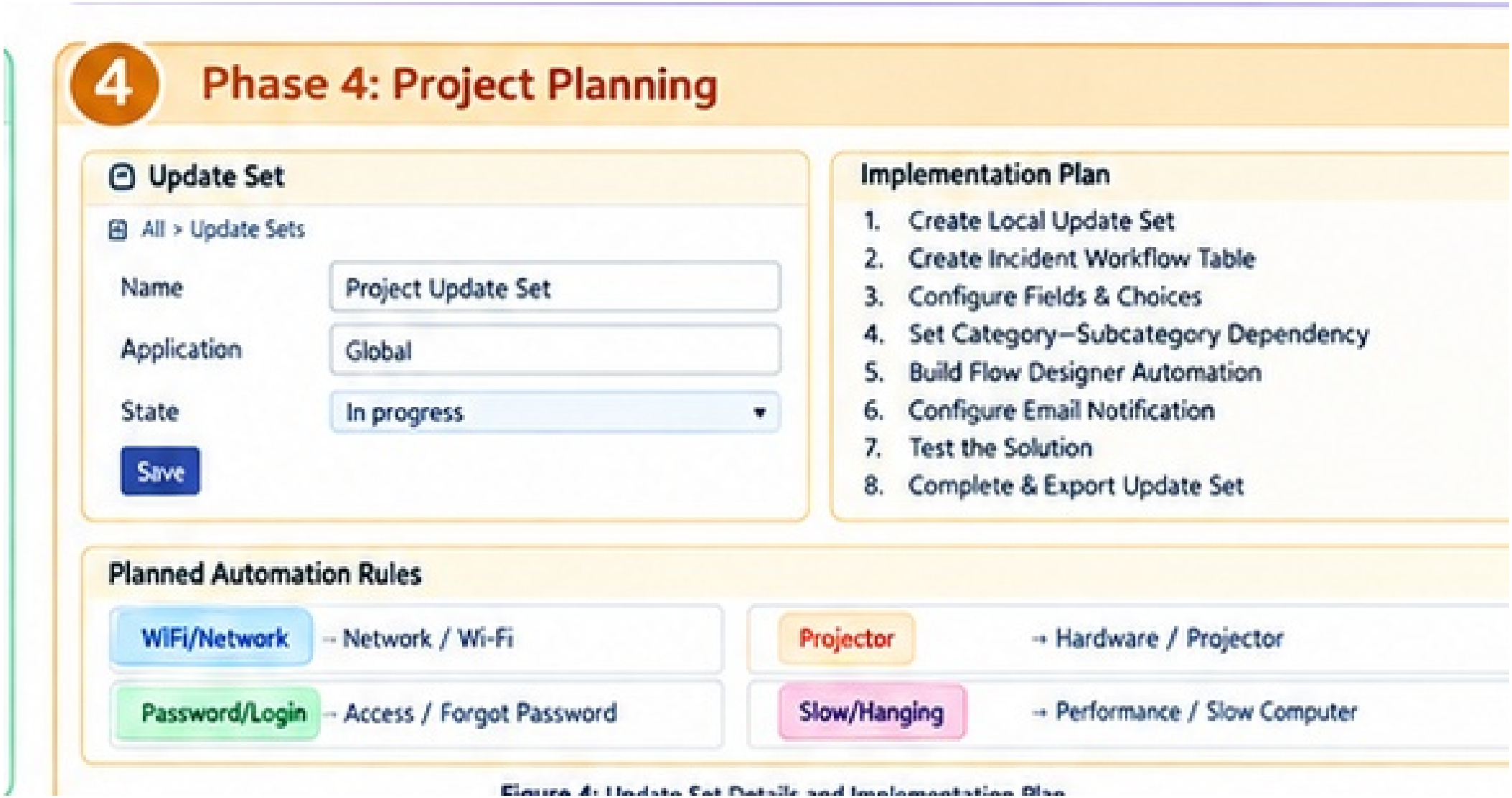
- Configured ServiceNow table and fields.
- Dependent choice configuration.
- Active Flow Designer flow.
- Automated email notification.
- Testing evidence and validation.
- Completed Update Set exported as XML.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the

short description/description fields and reduce manual effort.

Phase 4 · Illustrative Workflow / Configuration Diagram

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PHASE 5: Project Development

Objective: Implement the planned ServiceNow configuration and no-code automation.

1. Backend Configuration

- Create the Incident Workflow custom table under System Definition → Tables.
- Enable Auto Number for ticket records.
- Create the required fields with their appropriate data types.

2. Choice and Dependency Configuration

- Add choices to Category, Subcategory and State.
- Configure Subcategory as dependent on Category.
- Map each subcategory to its corresponding category.

3. Flow Designer Development

- Create a new Flow named Auto Classify School IT Tickets.
- Set the trigger to Record Created on the Incident Workflow table.
- Use the condition Category is Empty.
- Build If and Else If conditions using Short Description.
- Use Update Record actions to set Category and Subcategory.

4. Email Development

- Add a Send Email action after classification.
- Send the email to Caller → Email.
- Use the subject: Your Request for the issue has been submitted.
- Include a ticket confirmation message in the body.

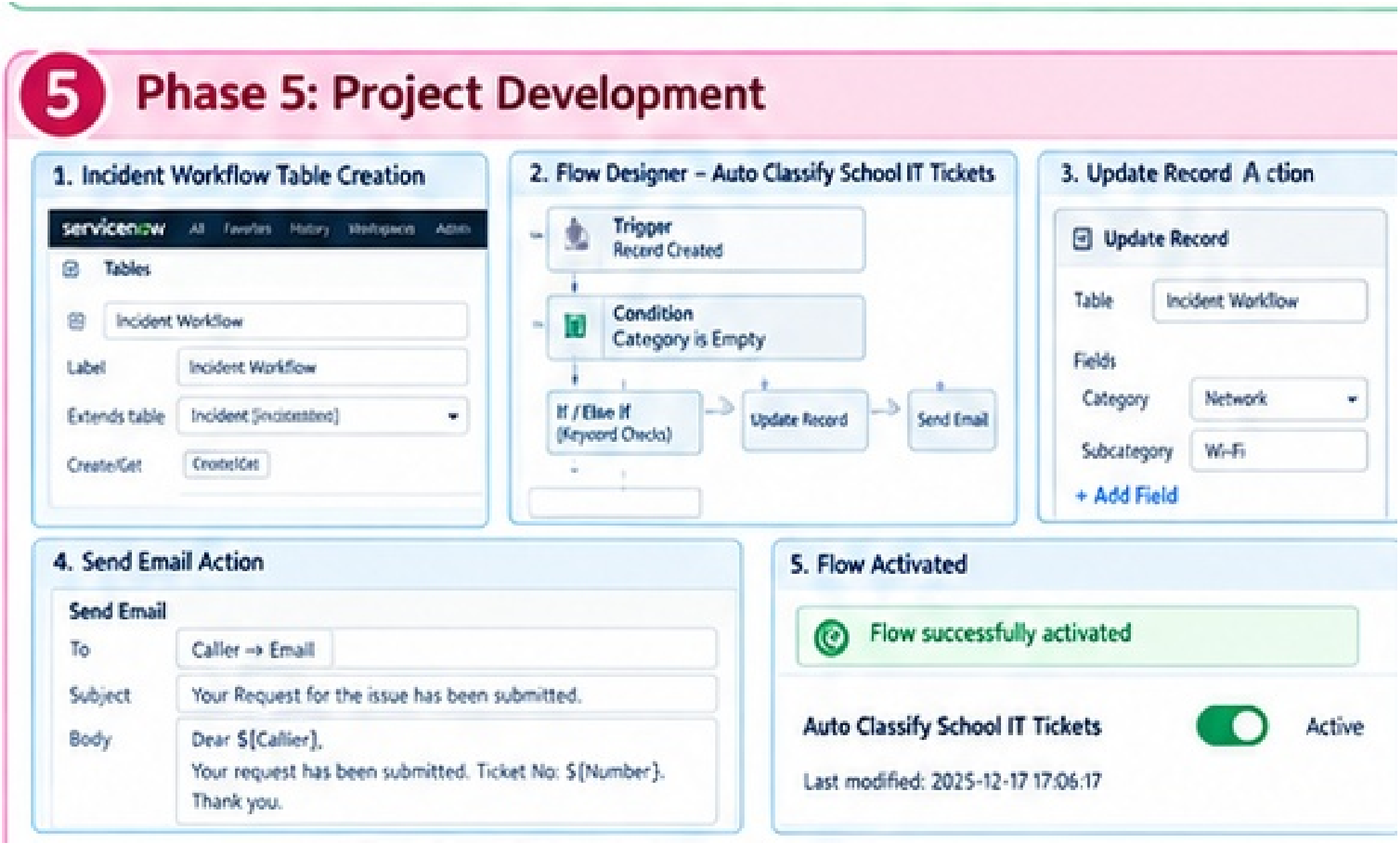
5. Activation

- Save and activate the completed Flow Designer flow.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 5 · Illustrative Workflow / Configuration Diagram

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PHASE 6: Project Testing

Objective: Validate the developed solution, confirm data accuracy and verify the end-to-end automation.

1. Validation Checks

- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to user and group tables.

2. Test Scenario – Wi-Fi

- Enter a Caller Name.
- Use Short Description: WiFi not working in library.
- Save and reload the form.
- Expected result: Category → Network; Subcategory → WiFi.
- Expected result: Email is sent to the caller email ID.

3. Test Scenario – Projector

- Enter a Caller Name.
- Use Short Description: Projector not turning on.
- Save and reload the form.
- Expected result: Category → Hardware; Subcategory → Projector.
- Expected result: Email is sent to the caller email ID.

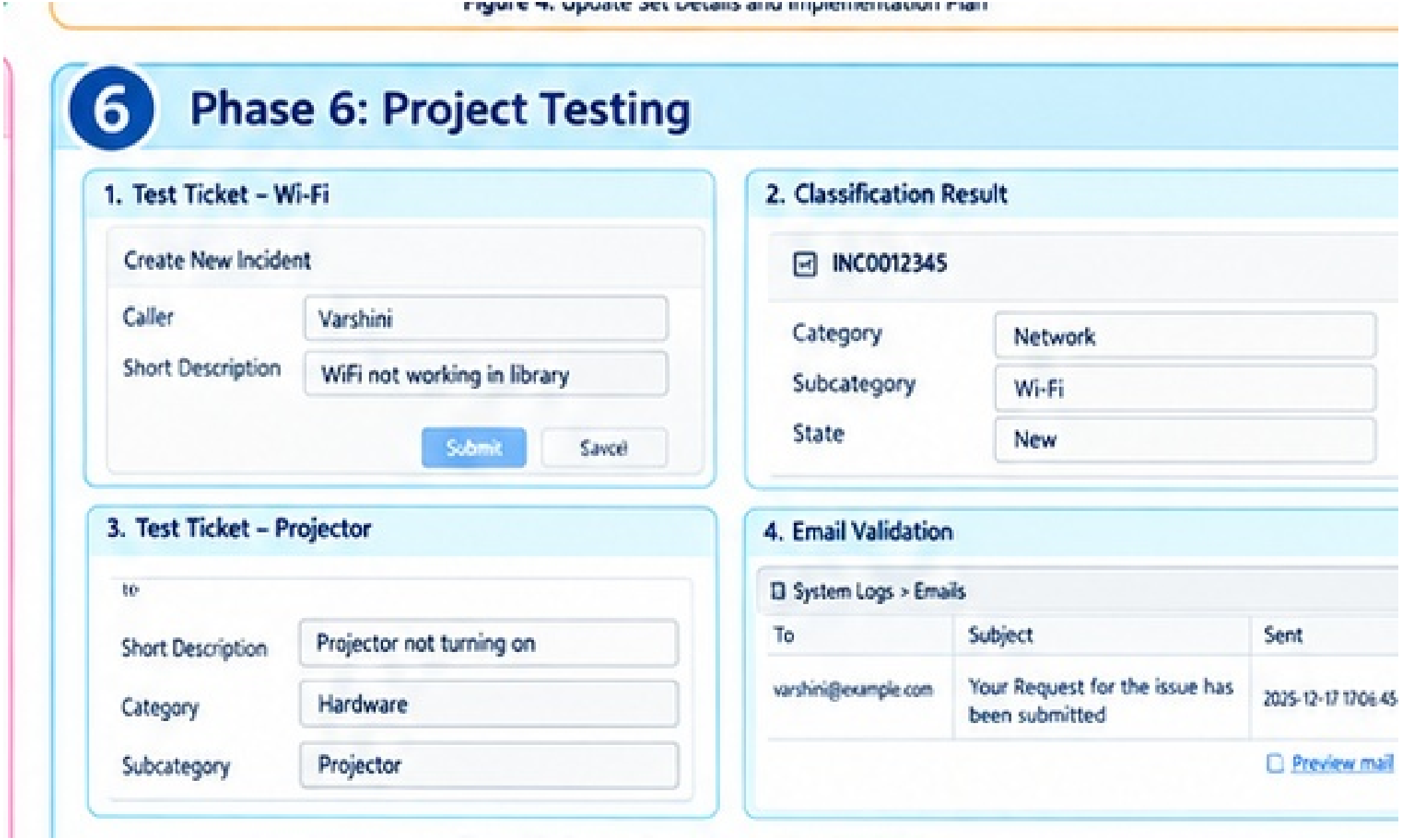
4. Email Validation

- Open All menus → Search Emails.
- Open System Logs → Emails.
- Search using the configured subject.
- Open the record and use Preview mail to verify the notification.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 6 · Illustrative Workflow / Configuration Diagram

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PHASE 7: Project Documentation

Objective: Prepare clear project records so the implemented solution can be understood, maintained and reused.

1. Documentation Coverage

- Document the problem statement and project objective.
- Document business requirements and required ticket fields.
- Document table creation, field configuration and choice values.
- Document the Category–Subcategory dependency mapping.
- Document Flow Designer trigger, conditions and update actions.
- Document email notification configuration.
- Document testing scenarios and expected results.
- Document deployment and Update Set export steps.

2. Update Set Documentation

- Open Local Update Sets and select Project Update Set.
- Change State from In progress to Complete.
- Save the form.
- Use Export to XML to download the update set data.
- The exported XML can be shared through WhatsApp or email and used for transfer/reuse.

3. Documentation Outcome

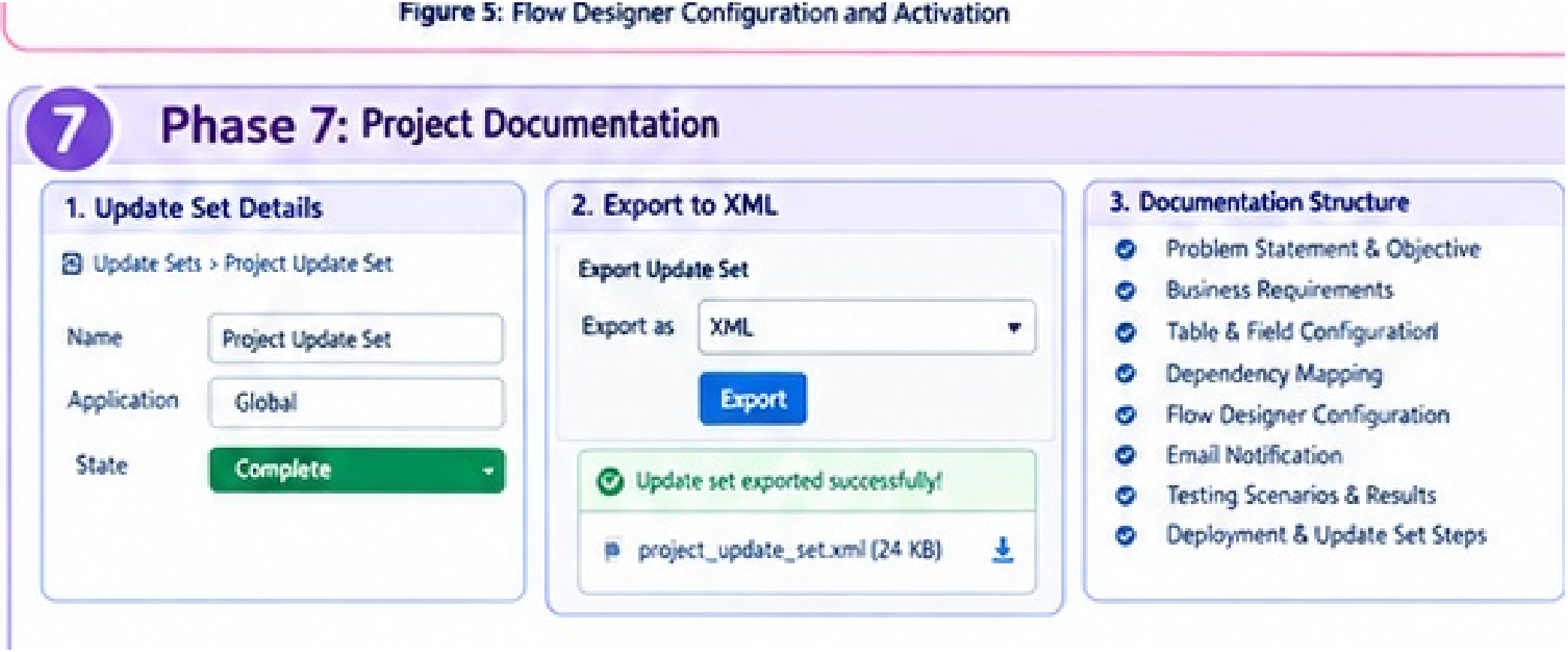
- The project has a structured record of requirements, design, development, testing and deployment.
- The documentation supports future maintenance and scalability.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 7 · Illustrative Workflow / Configuration Diagram

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Figure 5: Flow Designer Configuration and Activation



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PHASE 8: Project Demonstration

Objective: Demonstrate the complete working solution from ticket creation through automatic classification and email confirmation.

1. Demonstration Flow

- Create a new IT support ticket.
- Enter the Caller Name and Short Description.
- Submit the ticket without manually selecting Category or Subcategory.
- Show the Flow Designer automation triggering on record creation.
- Show automatic keyword-based classification.
- Show the resulting Category and Subcategory.
- Show the email notification sent to the caller.

2. Demonstration Example – Wi-Fi

- Input: WiFi not working in library.
- Automatic result: Category = Network; Subcategory = WiFi.
- Email confirmation is sent to the caller.

3. Demonstration Example – Projector

- Input: Projector not turning on.
- Automatic result: Category = Hardware; Subcategory = Projector.
- Email confirmation is sent to the caller.

4. Final Project Outcome

- The project provides end-to-end automation for school IT ticket classification.
- Manual categorization effort is reduced.
- Dependent choice fields improve data accuracy.
- Automated email notification improves communication.
- The no-code design is maintainable and scalable.
- The solution can be extended in future with assignment automation, SLA tracking or predictive intelligence.

Project Context: The school IT helpdesk receives requests such as Wi-Fi issues, projector failures, password problems and slow computers. The project uses ServiceNow Flow Designer to classify incidents from keywords in the short description/description fields and reduce manual effort.

Phase 8 · Illustrative Workflow / Configuration Diagram

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